Lecture 23 Scribe

L23_Amazone Sale_2026_...
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 23 slides).Instructions:Use 'One-Shot Prompting'. I will provide you with an Example (The Shot) of how to convert a slide concept into a scribe format. You must follow this structure and style to generate the scribe for the Lecture 23 slides.EXAMPLE (THE SHOT):Input Context: 'Slide 1: Definition of Set. A set is a collection of distinct objects. Example: A = {1, 2, 3}. Notation: x ∈ A means x is in set A.'Output Scribe:'### 1. Fundamentals of SetsDefinition: A set is a well-defined collection of distinct objects.Notation:$A = \{1, 2, 3\}$ represents a set with elements 1, 2, and 3.$x \in A$ denotes that element x belongs to set A.'YOUR TASK:Now, generate the full lecture scribe for the attached Lecture 23 slides following the style and faithfulness of the example above.Faithfulness: Do not add outside information. Use exact definitions/notations from slides.Structure: Use clear headings (e.g., Sample Space, Events, Probability Axioms) as per the slides.Goal: Exam-ready reference material.

Here is the **exam-ready lecture scribe** strictly based on the provided Lecture 23 slides , following your required style:

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1. Amazon Black Friday Sale 2026 – Context

The Amazon Black Friday Sale 2026 is expected to begin on **Friday, November 27, 2026**.

Expected Timeline:

- Early Deals: Starting as early as **November 20–21, 2026**
- Main Sale Period: **November 25–27, 2026**
- Cyber Monday: **November 30, 2026**

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2. Uncertainty in Computing Systems

Modern distributed systems (e.g., datacenters) experience **random job arrivals**.

Key Modeling Insight:

- Job arrivals can be modeled using **Poisson processes**
- Focus is on events like:
  - "Server has ≥ k jobs"
  - "≥ k arrivals within one hour"

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3. Tail Probability and System Risk

Definition:
Tail probability measures extreme events:

$$P(X \geq k)$$

Insight:

- The **mean behavior is safe**
- The **tail behavior is catastrophic**

Example (from simulation):

- Crash threshold: $k = 60$
- Need to evaluate:

$$P(X \geq 60)$$

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4. Need for Probability Bounds

Exact probabilities are often hard to compute.

Solution:
Use **upper bounds** on tail probabilities.

Hierarchy of Bounds:

1. **Markov Inequality**
   - Requires: Mean (μ)
2. **Chebyshev Inequality**
   - Requires: Mean (μ), Variance (σ^2)

3. **Chernoff Bound**
    - Requires: Full distribution (MGF)

Key Insight:
More information ⇒ **tighter bounds**

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5. Markov's Inequality

Definition:

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Properties:

- Uses only the mean
- Decay rate: **Linear** $(1/k)$
- Produces **loose (pessimistic) bounds**

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6. Chebyshev's Inequality

Definition:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Equivalent Tail Form:

$$P(X \geq a) \leq \frac{\sigma^2}{(a - \mu)^2}$$

Properties:

- Uses variance
- Decay rate: **Polynomial** $(1/k^2)$
- Tighter than Markov

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7. Chernoff Bound

Definition:

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

where:

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Properties:

- Uses Moment Generating Function (MGF)
- Decay rate: **Exponential** (e^{-k})
- Provides **tightest bounds**

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8. Comparison of Bounds (Decay Behavior)

- Markov: Linear decay → $1/k$
- Chebyshev: Polynomial decay → $1/k^2$
- Chernoff: Exponential decay → e^{-k}

Key Insight:
Chernoff closely matches real tail behavior.

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9. Case Study: Black Friday SLA Breach

Scenario Parameters:

- Average load: $\lambda = 500$ requests/min
- SLA threshold: $k = 600$

Objective:

$$\text{Compute } P(X \geq 600)$$

Goal:

- Predict failures
- Size backup capacity

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10. Cost of Loose Bounds

Observed Results:

- Exact risk: ~1.4%
- Chernoff: ~8.8%
- Chebyshev: ~20%
- Markov: ~90.9%

Insight:

Loose bounds ⇒ **massive overestimation**

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11. Financial Impact

Waste Multiplier:

$$\text{Waste} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

Example:

- Markov → extreme over-provisioning
- Chebyshev → moderate waste
- Chernoff → near optimal

Conclusion:

Bad math ⇒ **wasted infrastructure cost**

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12. Distribution Sensitivity

Different workloads follow different distributions:

- Normal → Stable jobs
- Poisson → Web requests
- Binomial → Fixed users
- Gamma → Bursty, heavy-tailed traffic

Insight:

- Chernoff works best with clean MGF
- For heavy-tailed cases → fallback to Chebyshev or simulations

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13. Engineer's Decision Matrix

Method	Requires	Decay	Tightness	Use Case
Markov	Mean (μ)	Linear $1/k$	Loose (10–1000x)	Quick sanity check
Chebyshev	Mean, Variance	Polynomial $1/k^2$	Moderate (2–10x)	Unknown distribution

Method	Requires	Decay	Tightness	Use Case
Chernoff	Full MGF	Exponential e^{-k}	Tight (<2x)	SLA-critical systems

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14. Class Exercise: Laplace Distribution

Given:

$$X \sim \text{Laplace}(\mu, b)$$

Objective:

- Derive Chernoff bound

Threshold:

$$a = \mu + 3\sigma$$

Reminder:

$$\sigma = b\sqrt{2}$$

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15. Derivation Blueprint

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2: Apply Chernoff

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3: Optimize

- Differentiate w.r.t t
- Set derivative = 0

Step 4: Substitute

- Plug optimal t into bound

Final Key Takeaways

- Tail events determine **system failures**
- Exact probabilities are hard ⇒ use bounds
- **Markov** < **Chebyshev** < **Chernoff** (in tightness)
- Better bounds ⇒ **better engineering decisions**
- Poor bounds ⇒ **financial over-provisioning**

If you want, I can convert this into a **1-page cheat sheet** or **derivation-focused notes** for exams.